

सेट / SET: 1
MARKING SCHEME
केंद्रीय विद्यालय संगठन ,जयपुर संभाग
KENDRIYA VIDYALAYA SANGATHAN, JAIPUR REGION
1st- PRE BOARD EXAMINATION :2024-25
कक्षा / CLASS: XII
विषय /SUB: PHYSICS (कोड / CODE:042)

Time Allowed: 3 Hours

Maximum Marks: 70

Q.No.	Model Answer	Marks
1	B	1
2	B	1
3	A	1
4	D	1
5	B	1
6	B	1
7	A	1
8	C	1
9	D	1
10	A	1
11	B	1
12	D	1
13	C	1
14	A	1
15	A	1
16	A	1
17	a) Microwave b) Infrared c) X-Ray d) UV Ray	1
	Microwave > Infrared > UV Ray > X-Ray	1
18	Correct Derivation of $I = neAV_d$	2
19	Correct formula of self-inductance	1
	Calculation of emf $e = 6.28 \times 10^{-4} \text{ V}$	1
20	Correct graph	1
	Correct derivation of slop of graph $= h/e$	1
	OR	
	Einstein photo electric equation	1/2
	i) Kinetic energy become more than double	

	ii) No change in photo current iii) No change in Work function	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
21	Diagram for N-type semi-conductor Diagram for P-type semi-conductor	$\frac{1}{2} + \frac{1}{2}$ $\frac{1}{2} + \frac{1}{2}$
22	$m({}_{26}\text{Fe}^{56}) + Q \rightarrow 2({}_{13}\text{Al}^{28})$ $\Delta m = 2 \times (27.98191)\text{u} - 55.934944\text{u}$ $= -0.02888\text{u}$ Energy needed = $0.02888 \times 931\text{MeV} = 26.88728\text{MeV}$	1 1 1
23	Correct diagram of parallel plate capacitor with dielectric slab Derivation of capacitance Effect of metal sheet (Dielectric constant of metal is infinite)	1 $1 + \frac{1}{2}$ $\frac{1}{2}$
24	Three equations using Kirchhoff's law's Calculation of current $I_1 = 2.5\text{A}$ $I_2 = 5/8\text{A}$ $I_3 = 15/8\text{A}$	$1 + \frac{1}{2}$ $1 + \frac{1}{2}$
25	Diagram of refraction using Huygens's principle Derivation of Snell's law	1 2
26	Bohr Postulates Proof of circumference = $n\lambda$ OR Rydberg relation for wavelength Longest wavelength = 655.9nm Shortest wavelength = 364.4nm	1 2 1 1 1
27	Formula for fringe width ($\lambda D/d$) Calculation of wavelength $\lambda = 600\text{nm}$	1 2
28	Difference between Para ,Ferro and Dia magnetic substances Example of Para ,Ferro and Dia magnetic substances	$\frac{1}{2}, \frac{1}{2}, \frac{1}{2},$ $\frac{1}{2}, \frac{1}{2}, \frac{1}{2},$

29	i)B ii) C iii) C iv) A OR C	1x4=4
30	i)A ii) D iii) A iv) C OR D	1x4=4
31	a) Phasor diagram Derivation of impedance Resonance b) Formula of resonant frequency New resonant frequency =2f OR a) Principle Labeled diagram Working b) Formula for efficiency $I_P = 15A$ $E_S = 450V$	1 1 1 1 1 1 1 1 ½ ½ 1
32	a) Circuit Diagram Working Input- Output wave form b) Total resistance =360ohm Current I= (1/60)A OR a) Circuit diagram Working Input- Output wave form b) Identification of semiconductor X= P- type Y= N-type i) Reverse Bias ii) V-I graph of reverse Bias	1 1 1 1 1 1 1 1 ½ ½ ½ ½
33	a) Gauss's theorem Diagram Derivation of electric field $E = 2(K\lambda / r)$ b) i) $5.4 \times 10^6 N/C$ ii) $8.1 \times 10^{-3} N$ OR a) Definition and unit of flux Diagram Derivation of electric field due to plane sheet b) i) $-10^3 Nm^2/C$ ii) $-8.8nC$	1 1 1 1 1 1 1 1 1 1

